

AsimovBM: Toward Automatic Evaluation of Human-Perceived Traits in Physical AI Robots

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Abstract—Physical AI robots entering human-centered environments are judged not only by whether they finish a task, but by how their behavior feels to the people around them. A policy may reach the goal and still appear unsafe, illegible, or socially inappropriate, echoing the practical concerns behind Asimov’s laws of robotics; standard engineering metrics rarely capture this human experience. We present *AsimovBM*, a simulation-based benchmark that automatically estimates four human-perception macro-scores (*Perceived Dexterity*, *Perceived Safety*, *Perceived Social Awareness*, and *Impression*) from objective behavioral telemetry, reducing the need for per-configuration user studies. A client–server design keeps the evaluated policy on the submitter’s hardware while hidden scenarios run on the server, simultaneously protecting intellectual property and reducing benchmark overfitting. Preliminary validation on two Unitree G1 humanoid policies, a standard navigation policy and an aggressive-speed variant, shows that automated scores closely track human judgments: a per-axis linear alignment fitted on 50 in-sample participants achieves 91.6% policy-gap agreement with human raters, and generalises to 70 held-out participants, yielding the expected ranking between the two policies across all four human-perceived metric axes. These results suggest that human-perceived robot behavior can be evaluated automatically, reproducibly, and without exposing the policy under test, while raising a hypothesis: interpretable assumptions about motion, proxemics, and interaction geometry can predict human preference well enough to guide human-aligned policy improvement.

Index Terms—Human-Robot Interaction, Human Perception, Physical AI Benchmark, Automatic Evaluation, Social Navigation

I. INTRODUCTION

A robot operating around people must complete its task while appearing safe and predictable; these perceptions affect acceptance, trust, and cooperation [1]–[4].

Task success, path length, collisions, and completion time can be measured automatically in simulation. Human-facing qualities are usually assessed with instruments such as Godspeed [5], NARS [6], and RoSAS [7]. Because these instruments require participants, repeated evaluation during policy development is impractical, and studies are often limited to a final policy whose results may not transfer across robots, tasks, or populations.

Robot trajectories contain several cues that affect human ratings: clearance from people, speed during close passes, orientation toward the target person, and commitment to a direction. AsimovBM computes sixteen such features from

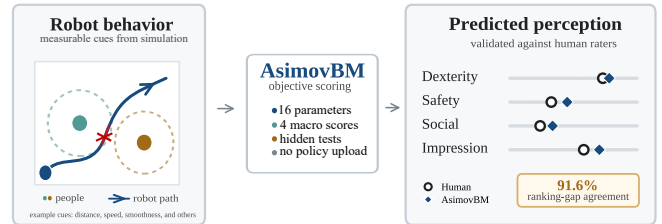


Fig. 1. Overview of AsimovBM. In the proposed client–server design, the server runs the scenario and records telemetry while the policy remains on the submitter’s hardware. The server converts telemetry into sixteen features and four human-facing scores for comparison with human ratings.

simulation telemetry and combines them into four scores aligned with the study questions (Fig. 1).

The mapping is not learned from the study data; its weight matrix follows prior work on proxemics, legibility, and robot perception. Each score decomposes into named features with explicit thresholds, so a low score can be traced to a specific behavior, such as a fast pass within 0.45 m of a bystander. This decomposition also identifies the source of the largest social-awareness discrepancy in Section V.

To avoid requiring proprietary policy uploads, the proposed client–server design places the simulation on the benchmark server and the policy on the submitter’s hardware, with observations and actions exchanged through a versioned protocol. The design does not disclose scenario parameters, which reduces direct access to the evaluation set and the risk of test-set overfitting [8], [9].

We evaluate the metric framework with 120 participants who rated two Unitree G1 locomotion policies across three social-navigation scenarios. The automated scores align with human judgments across all four axes, with the strongest agreement on dexterity, where the human difference is also clearest. Appendix A identifies the features behind the larger discrepancy on social awareness. Per-axis affine alignment maps automated scores to the human response scale, but an alignment based on two policies cannot demonstrate predictive validity (Section V-B).

This paper contributes:

- a metric framework that maps sixteen telemetry features to four human-facing axes using documented thresholds, a literature-grounded weight matrix, explicit feature-level

handling of non-applicability and insufficient evidence, and hard caps for safety and task failures;

- the design of a client–server architecture that keeps policy weights with the submitter and scenario parameters on the benchmark server; and
- a study of two contrasting G1 policies with 120 participants, including the limits of two-policy calibration and analyses of camera viewpoint and prior robot experience.

II. RELATED WORK

Questionnaires are validated and widely used tools for measuring robot perception. Godspeed [5], [10] measures anthropomorphism, animacy, likeability, perceived intelligence, and perceived safety; NARS [6], [11], [12] measures negative attitudes toward robots; and RoSAS [7] separates warmth, competence, and discomfort. Studies of anthropomorphism, proxemics, and approach distance identify spacing, approach angle, speed, and appearance as relevant physical cues [13]–[17]. These cues motivate several AsimovBM features, but their meaning depends on context, so a questionnaire score alone does not define socially appropriate behavior [18]–[20]. Unlike work that infers human properties from interaction cues, AsimovBM evaluates robot behavior [21].

Social-navigation research reports objective measures such as minimum distance, personal-space intrusion, path efficiency, time-to-collision, and jerk [22]–[27]. Guidelines and simulation benchmarks improve comparability [28]–[30], but do not specify how a value such as 0.7 m of minimum clearance maps to perceived safety. Adaptive proxemics and explicit robot–human signaling show that this mapping depends on context [4], [31]. AsimovBM retains these established physical measures and adds an explicit mapping to four perceptual axes.

Other work predicts perception or preference directly from behavior. Learned evaluators predict perceived navigation performance, evaluate autonomous HRI behavior, or relate competence and legibility to corrective human feedback [32]–[34]. Preference-based methods learn rewards from human or AI feedback [35]–[40], and planners can account for how robot actions affect human behavior [41]. These methods can capture relations missed by fixed weights, but require training data and provide less direct attribution. AsimovBM uses human ratings to evaluate and scale the mapping, not to fit its weights. Dialogue and apparent personality also affect perception but remain outside a telemetry-only benchmark [42].

Public benchmarks are vulnerable to direct optimization against their test sets. HELM documents overfitting to public evaluations [8], while Chatbot Arena uses hidden inputs and pairwise comparisons to improve ranking robustness [9]. Because robotics evaluation must also protect proprietary policies, the proposed AsimovBM client–server design keeps scenarios on the server and policy weights with their owner. It also links questionnaire constructs to automated social-navigation measures without learning an opaque evaluator. The fixed mapping trades flexibility for feature attribution.

III. ASIMOVBM FRAMEWORK

Each episode produces a trace of robot pose and velocity, control commands, human positions, and structured task events. From this trace, AsimovBM computes four scores in $[0, 100]$: perceived dexterity ($\mathcal{D}$), perceived safety ($\mathcal{S}$), perceived social awareness ($\mathcal{A}$), and impression ($\mathcal{I}$). These axes match the four study items and established constructs: competence, physical and psychological safety, social appropriateness, and overall attitude. The study compares benchmark and participant rankings across these axes.

A. Metric Definition and Aggregation

Every feature x is mapped to a score $R \in [0, 100]$ by linear interpolation between documented good and bad reference values:

$$R = 100 \cdot \begin{cases} \text{clip}\left(\frac{x - x_{\text{bad}}}{x_{\text{good}} - x_{\text{bad}}}, 0, 1\right), & \text{higher is better,} \\ \text{clip}\left(\frac{x_{\text{bad}} - x}{x_{\text{bad}} - x_{\text{good}}}, 0, 1\right), & \text{lower is better.} \end{cases} \quad (1)$$

The thresholds encode the domain assumptions and are stored in versioned scenario configuration, so adapting the benchmark to a different robot, culture, or study population requires new configuration values rather than new formulas. Minimum clearance, for example, is anchored at Hall’s personal-space boundary $d_{\text{ps}} = 1.2$ m on the good side and at a hard floor $d_{\text{hard}} = 0.45$ m on the bad side [14]; motion smoothness is anchored at SPARC values of -2.0 and -6.0 [43].

Table I lists all sixteen features, groups them by their main axis, and gives their normalization thresholds. Dexterity is built from task outcome and path quality, safety from proxemic geometry, social awareness from responses to structured social cues, and impression from motion quality. The full definitions, together with the applicability rules, are in Appendix A.

TABLE I
THE SIXTEEN BEHAVIORAL FEATURES AND THRESHOLDS OF EQ. (1).

Feature	Good	Bad
<i>Perceived Dexterity ($\mathcal{D}$)</i>		
Task success rate	1.0	0.0
Task completion time	$0.35 T_{\text{max}}$	T_{max}
Comfort-aware path efficiency	1.0	0.35
Hesitation rate	0 ev/s	0.5 ev/s
<i>Perceived Safety ($\mathcal{S}$)</i>		
Minimum human-robot distance	1.2 m	0.45 m
Proxemic intrusion dose	0	$0.25 d_{\text{ps}} T$
Speed near humans (P95)	0.25 m s^{-1}	1.0 m s^{-1}
<i>Perceived Social Awareness ($\mathcal{A}$)</i>		
Gesture response success	1	0
Acknowledgment clarity	1	0
Human-aware approach	0.9	0.4
Bystander acknowledgment	0.5	0.05
<i>Impression ($\mathcal{I}$)</i>		
Motion smoothness (SPARC)	-2.0	-6.0
Heading jerk	-2.0	-6.0
Stability	1.0	0.0
Legibility (commit time)	0.7	0.1
Behavioral naturalness	1.0	morphology-specific

A feature that the scenario does not exercise is marked *not applicable* and excluded from the aggregate; for example, human-clearance features are not applicable in an empty room. A feature with too few valid observations is marked *insufficient evidence*, and an axis is suppressed entirely when a tier yields fewer than three valid episodes.

Aggregation and caps. Features contribute to axes through an evidence-weight matrix $W \in \mathbb{R}_{\geq 0}^{16 \times 4}$. A feature may load on several axes because the perceptual constructs overlap: instability can reduce both perceived safety and overall impression. Each entry was assigned on a 0–3 scale from literature on that behavior’s perceptual effect, drawing on robot-perception and attitude instruments [5], [6], proxemics and human-aware navigation [14], [26], [27], and legibility and smoothness [43], [44]; Fig. 2 shows the resulting matrix, column-normalized.

The uncapped axis score is the weighted mean over the active features,

$$S_k^{(0)} = \frac{\sum_{i \in \mathcal{A}_k} w_{ik} R_i}{\sum_{i \in \mathcal{A}_k} w_{ik}}, \quad (2)$$

where $\mathcal{A}_k$ holds the scored features with nonzero weight on axis k ; dropping a feature renormalizes the rest.

To prevent smooth motion from compensating for failures, AsimovBM applies hard caps after Eq. (2): the reported score is $S_k = \min(S_k^{(0)}, C_k)$ when a cap C_k is triggered. A breach of the hard distance floor caps safety at 25; a collision, fall, or invalid action caps it at 20; a fast close pass, meaning both proxemic dose and near-human speed in their bad ranges, caps it at 35. A task success rate of zero caps dexterity at 35, and when several caps fire, the lowest applies. These caps give both policies a safety score of 20.0 in the moving-bystander scenario of Section V, making that condition uninformative.

B. Alignment to the Human Rating Scale

The axis scores and rescaled Likert responses both lie in $[0, 100]$, but their numerical levels are not directly equivalent. To compare them we fit one affine map per axis,

$$\hat{s}_k = \text{clip}_{[0,100]}(a_k + b_k S_k), \quad (3)$$

on a calibration split of participants, and report $\hat{s}_k$ against the ratings of the held-out participants. The composite reported in the results is the unweighted mean of the four aligned axis scores. The operational benchmark report uses the same equal-axis mean and applies a global safety cap when a hard safety event occurs. With only two policies, this procedure demonstrates scale alignment, not predictive accuracy (Section V-B).

C. Client–Server Architecture Design

The metric framework is independent of the episode-generation system. In the proposed client–server design, policy weights stay with the submitter and evaluation scenarios stay private to limit direct optimization against the test set.

In this design, the server runs the MuJoCo simulation [45], samples bystander placement and motion patterns, and streams named sensor observations and structured task events to the policy, which runs on the submitter’s hardware and returns

actions. Submitted policy code does not execute on the server, and scenario configuration is not sent to the client. Metric inputs are computed from server-owned telemetry, so a policy cannot read privileged simulator state or report an inflated score. The same interface supports wheeled bases, legged robots, and mobile manipulators; morphology is specified through the robot package and morphology-specific features.

The protocol separates technical from behavioral outcomes by reporting setup errors, package rejections, policy exceptions, invalid actions, timeouts, and disconnections separately from behavioral scores and from the *not applicable* and *insufficient evidence* states defined in Section III.

The local MuJoCo episode runner, metric computation, and reporting pipeline are implemented and tested. Client–server integration remains a design and will be implemented as future work; it was not used in these experiments.

IV. EXPERIMENTAL SETUP

Platform and policies. We simulate a Unitree G1 humanoid in MuJoCo and evaluate two locomotion policies selected to produce contrasting behavior: Policy A, the standard Unitree controller, and Policy B, a G1-ASAP variant that moves faster and yields less. This pair tests whether the metrics distinguish contrasting behavior but cannot establish generalization.

Scenarios. Each policy runs three social-navigation scenarios of increasing difficulty: an empty room, a room with static bystanders, and a room with moving bystanders. Each scenario contributes one behavioral episode per camera viewpoint, so six scored runs per policy and twelve in total. Only the third scenario places people on the robot’s path and contributes direct proxemic safety evidence.

Evaluation videos. The study used four rendered videos, one for each policy–viewpoint combination. The first-person view approximates the perspective of the person approached by the robot; the third-person view represents a bystander. The two viewpoints use separate runs of the same scenario configuration. Viewpoint changes the visual evidence available to participants, while automated scores depend only on each run’s telemetry (Section VI).

Human study. We recruited 120 participants through Politecnico di Milano and the AEA student association. Each participant viewed all four stimuli in the same order and rated each on four 7-point Likert items phrased in everyday language [46]: competence and control ($\mathcal{D}$), safety around people and obstacles ($\mathcal{S}$), awareness of the surroundings ($\mathcal{A}$), and overall impression ($\mathcal{I}$). The survey also recorded whether participants interact with robots daily and was presented in English and Italian. Participation was voluntary, anonymous, and uncompensated; no personal data beyond the familiarity item were recorded. Because each participant rated both policies, policy comparisons are paired; ratings are mapped linearly to $[0, 100]$, and Appendix B documents the full instrument.

Analysis. Participants were split into a calibration set of 50, used only to fit the affine maps of Eq. (3), and a held-out set of 70, used for the primary comparisons. The split was performed by participant, so each participant contributed to one set

How objective sub-metrics flow into human-perception scores

Sub-metric and interpretation		Dexterity	Safety	Social Awareness	Impression
Task performance	Task Success Rate acknowledges and stops at target	16%	4%	—	5%
	Task Completion Time time needed after success	5%	—	—	—
	Comfort-Aware Path Efficiency efficient approach outside comfort zones	5%	4%	2%	—
	Hesitation Rate stops, reversals, or indecisive motion	11%	6%	2%	5%
Human spacing	Minimum Human-Robot Distance closest instant to a person	—	11%	9%	3%
	Proxemic Intrusion Dose depth and duration inside personal space	—	11%	14%	11%
	Speed Near Humans fast close passes near people	5%	11%	5%	3%
Social response	Gesture Response Success responds correctly to an explicit cue	5%	7%	14%	11%
	Acknowledgement Clarity orients toward the intended target	5%	7%	14%	11%
	Human-Aware Approach progress, clearance, and approach angle	5%	11%	14%	11%
	Bystander Acknowledgement preserves clearance and slows near bystanders	5%	7%	14%	5%
Motion quality	Motion Smoothness (SPARC) smooth translational motion	5%	2%	—	5%
	Heading Jerk smooth turning behavior	5%	4%	—	8%
	Stability falls, collisions, and invalid actions	11%	11%	—	5%
	Legibility early commitment to an interpretable trajectory	11%	4%	14%	5%
	Behavioral Naturalness gait or wheel-velocity regularity	5%	—	—	11%

Dash = zero weight; darker cell = stronger influence

5% 10% 15%

Fig. 2. The evidence-weight matrix W maps sixteen features to four perceptual axes. Entries are literature-based 0–3 priors [5], [6], [14], [26], [27], [43], [44], shown as column-normalized percentages. Dashes denote zero weight, values below 3% are unlabeled, and each score can be traced to its features.

only. Human policy differences were evaluated with two-sided paired t -tests and bootstrap 95% confidence intervals. The reported p values are exploratory and unadjusted.

Agreement measure. Per axis we report the *gap match*,

$$\Delta\% = 100 - \left| (\hat{s}_{k,A} - \hat{s}_{k,B}) - (H_{k,A} - H_{k,B}) \right|, \quad (4)$$

which compares the automated and human A-versus-B gaps on the 0–100 scale.

V. RESULTS

A. Policy-Level Results

Automated scores. Table II reports the uncalibrated axis scores per scenario, before scale alignment with human ratings. Policy A scores at least as high as Policy B on every axis in every scenario, and strictly higher in eleven of the twelve comparisons; the exception is safety under moving bystanders, where both are capped at the same value. The dexterity gap widens steadily, from 7.9 points in the empty room to 22.4 with static bystanders and 33.2 with moving ones, because Policy B times out in the last scenario, which sets its task success rate to zero and caps its dexterity at 35. On social

awareness, the gap narrows from 43.6 to 12.9 because both policies receive lower scores in the moving-bystander scenario.

The safety scores require additional context because, in the moving-bystander scenario, both policies score exactly 20.0 after breaching the hard distance floor and receiving the same cap, leaving the saturated axis unable to discriminate between them. In the two scenarios without people on the path, the proxemic features are not applicable, so safety is computed only from cross-axis features such as stability and therefore measures little proxemic behavior in this experiment.

Human ratings. For the 70 held-out participants, the paired differences $D = A - B$ are in Table III and Fig. 3. Participants rated Policy A higher on dexterity ($D = 17.7$, $p < 0.001$) and impression ($D = 7.7$, $p = 0.002$). They did not distinguish the policies on safety ($D = 1.1$, $p = 0.735$), while the social-awareness mean slightly favors Policy B ($D = -4.3$) without reaching significance ($p = 0.152$).

With standard deviations of 20–26 points on the 0–100 scale, between-participant variation exceeds the mean policy difference on safety, social awareness, and impression; only dexterity has a large difference relative to this variation.

TABLE II
UNCALIBRATED AUTOMATED AXIS SCORES PER SCENARIO AND POLICY.

Scenario	Policy	Dext.	Safety	Social	Impr.
Empty room	A	92.9	90.9	100.0	82.4
	B	85.0	71.8	56.4	58.8
Static bystanders	A	91.7	89.4	98.8	82.4
	B	69.3	20.0	55.4	46.4
Moving bystanders	A	68.2	20.0	51.9	51.1
	B	35.0	20.0	39.0	32.6
All	A	84.3	66.8	83.6	71.9
	B	63.1	37.3	50.3	45.9

Higher is better. Scenario scores are averaged over the two camera viewpoints; the All row averages the three scenarios. In the moving-bystander scenario, safety is capped at 20 for both policies and Policy B times out, making the safety comparison uninformative.

TABLE III
PAIRED HELD-OUT RATINGS, POLICY A MINUS POLICY B.

Axis	Mean D	SD	p value
Dexterity	17.7	23.1	< 0.001
Safety	1.1	26.4	0.735
Social awareness	-4.3	24.7	0.152
Impression	7.7	20.5	0.002

$D = A - B$ on the $[0, 100]$ scale; two-sided paired t -tests, $n = 70$.

B. Agreement and Calibration Limits

Aligned scores. Comparing Table II with Table III shows broad agreement across all four axes before fitting. The uncalibrated gap magnitudes differ: the composite gap is 27.5 points for the benchmark and 5.5 points for participants.

Table IV reports the aligned scores from Eq. (3) on the held-out participants. The automated and human policy gaps then differ by 8.5 points on the composite, giving a gap match of 91.6%; per-axis values, shown in Fig. 4, range from 95.6% on dexterity to 88.6% on social awareness.

On the aligned scale, the automated composite scores are 47.7 for Policy A and 33.7 for Policy B. The largest deviation occurs for Policy B on social awareness, where the automated score of 31.4 lies just below the human 95% confidence interval, whose lower bound is 33.0. A more complex nonlinear alignment could reduce such residual differences, but we use a linear alignment in this study to limit overfitting.

TABLE IV
HELD-OUT COMPARISON OF HUMAN AND ALIGNED AUTOMATED SCORES.

Axis	Policy A		Policy B		Gap match
	Human	Auto	Human	Auto	
Dexterity	54.2 $\pm$ 5.2	56.2	36.4 $\pm$ 4.6	34.0	95.6
Safety	38.0 $\pm$ 5.4	43.0	36.9 $\pm$ 5.4	33.2	91.2
Social aware.	34.5 $\pm$ 4.9	38.4	38.8 $\pm$ 5.9	31.4	88.6
Impression	48.2 $\pm$ 4.9	53.1	40.5 $\pm$ 4.9	36.2	90.8
Composite	43.7 $\pm$ 4.6	47.7	38.2 $\pm$ 4.7	33.7	91.6

Human values ($n = 70$) are means with bootstrap 95% confidence-interval half-widths. Automated values are $\hat{s}_k$ from Eq. (3), fitted on 50 calibration participants. Gap match is Eq. (4); Section V-B gives its identifiability limit.

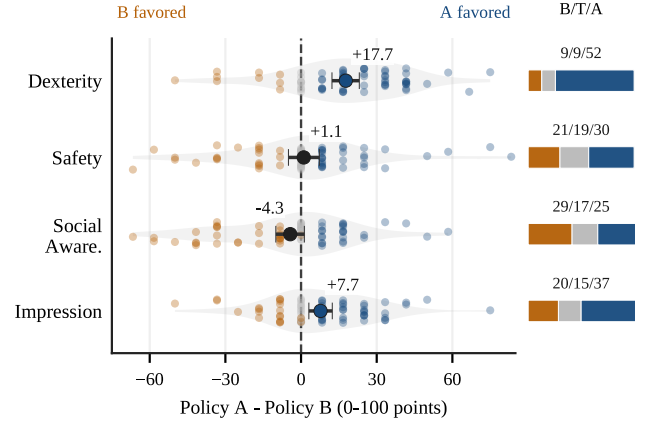


Fig. 3. Participant-level rating differences, Policy A minus Policy B, on the held-out set ($n = 70$). Small dots are individual participants, large markers and bars are means with bootstrap 95% confidence intervals. Except on dexterity, variation between raters exceeds the difference between policies.

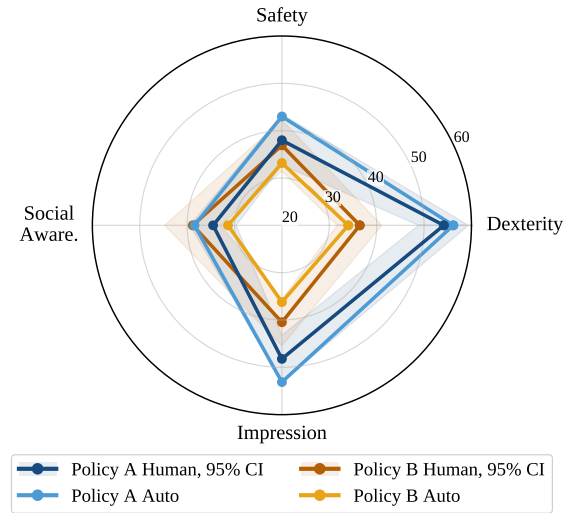


Fig. 4. Held-out human means and aligned automated scores for both policies on the four axes. Error bars are bootstrap 95% confidence intervals on the human means. Social awareness has the largest mismatch.

Calibration limit. The alignment of Eq. (3) has two free parameters per axis and is fitted to two automated scores, one per policy. The system is exactly identified: if the scores differ, the fitted map reproduces both calibration means. Table IV thus compares mean ratings between the calibration and held-out cohorts. Its gap-match values measure the stability of the human policy difference across the participant split, not the benchmark's predictive accuracy.

The aligned scores show good agreement across all four axes, while the feature values in Appendix A explain why social awareness has the largest discrepancy. Testing predictive generalization requires more than two policies, so that the affine map has more observations than parameters and leaves a measurable residual.

VI. DISCUSSION

Agreement on dexterity. Dexterity has the clearest human difference and the closest agreement with the benchmark because task success, completion time, and path efficiency are measured directly in simulation and visible to viewers, with timeouts evident in both telemetry and video.

Social-awareness alignment. Social awareness has the largest discrepancy, but its gap match remains high at 88.6%. The feature values in Appendix A trace most of this difference to acknowledgment clarity: Policy B fails the feature in five of six runs because it approaches without first turning toward the caller, and this feature carries a large social-awareness weight. Participants may instead have interpreted the immediate motion as responsiveness.

Saturation of the safety axis. Only one of three scenarios put people on the robot’s path, and both policies breached the hard distance floor there, received the same cap, and showed no difference in participant ratings ($p = 0.735$). Because no condition had one policy respect personal space while the other violated it, future scenarios should include proxemic interactions that do not immediately trigger the caps.

Viewpoint dependence. The automated score is camera-invariant because it uses telemetry, but human ratings can change with viewpoint. The third-person view widened the policy gap on all four axes, most clearly on safety (+0.614 Likert points, $p = 0.021$) and social awareness (+0.652, $p = 0.014$), without shifting the overall rating level. These exploratory, uncorrected tests suggest that human studies should control and report viewpoint.

Rater dependence. Participants with daily robot contact ($n = 23$) rated the policies 9.0 points lower on average than participants without daily contact ($n = 95$), with a bootstrap 95% interval of $[-16.5, -1.3]$ and the largest difference on social awareness (-10.0). This association does not establish causality but suggests that ratings differ across rater populations (Appendix B).

VII. LIMITATIONS AND FUTURE WORK

Validation scale. The study covers one robot, two policies, three scenarios, and one episode per scenario and viewpoint. The gap-match results show good agreement across all four axes, although the validation set remains small. Each scenario has two viewpoint-specific runs, below the framework’s minimum of three valid episodes for a tier; the per-scenario values in Table II are individual measurements, not tier estimates. With two policies, the affine alignment is exactly identified (Section V-B). Validation with six to ten policies and at least three valid repetitions per condition would provide a held-out gap error and support per-axis correlation analysis.

Metric and scenario coverage. The weights in W come from prior work and were not fitted to these data. Until they are tested across more policies, they remain documented modeling choices. The social-awareness result indicates that the acknowledgment-clarity weight requires further study. Heading jerk reaches the bad threshold in all twelve runs and

contributes only a constant; its signal or threshold must be re-derived for this platform. In addition, only one scenario places people on the robot’s path, where both policies immediately trigger the safety cap, so weight revisions require more varied scenarios and feature ablations.

Study validity. All behavior is simulated and shown through video, which cannot reproduce physical co-presence. Participants also saw the stimuli in a fixed order, and the introduction named the paper and its evaluation goal, so order effects and demand characteristics cannot be excluded. Real-world studies should counterbalance stimulus order and use a neutral description. Recruitment centered on one academic network, English and Italian were shown together. Proxemic expectations can vary across cultures, so the alignment should be treated as population-specific, consistent with the rater-experience result in Appendix B.

Deployment and scope. The local simulation, metric computation, and reporting pipeline are implemented, while client-server integration remains a design and will be implemented as future work. No external team has yet submitted a policy through this architecture, so its behavior in external use has not been validated. The current telemetry omits interaction timing, doorway yielding and negotiation, recovery after social errors, ambiguous human motion [47], speech, personality, and dialogue style [42]; these behaviors require new scenarios and observation channels.

VIII. CONCLUSIONS

AsimovBM provides repeatable estimates of four human-facing qualities by combining sixteen robot-telemetry features through a documented, hand-specified weight matrix. Each score decomposes into named features with explicit thresholds, making human-rating disagreements traceable to assumptions.

Across two Unitree G1 policies and 120 participants, the scores align with human judgments across all four axes, with the strongest agreement on dexterity, which also has the clearest human difference. Social awareness has the largest remaining discrepancy. Affine alignment places scores on the human scale, but two policies are insufficient to demonstrate predictive accuracy.

Validation of the affine maps requires more than two policies because each map has two fitted parameters. It must also cover proxemic behavior without immediately saturating safety caps, reassess social-awareness features in targeted studies, and evaluate external submissions on hidden scenarios. Automated estimates can guide development alongside human validation.

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APPENDIX A

METRIC DETAILS AND FEATURE-LEVEL RESULTS

This appendix defines the sixteen behavioral features and reports their values for both policies after normalization by Eq. (1) using the stated good and bad thresholds. Let $\mathcal{E}$ be the valid behavioral episodes of a tier and $\mathcal{E}_s \subseteq \mathcal{E}$ the successful ones, where success means the robot acknowledges the target, reaches it, and stops within the band $[0.8, 1.4]$ m before the timeout $T_{\max} = 120$ s. An axis is reported as *insufficient evidence* when $|\mathcal{E}| < 3$.

Perceived dexterity. Perceived dexterity measures how competently the robot executes the navigation task. Task success and hesitation rates use all valid episodes; completion time and path efficiency use successful episodes only.

Task Success Rate (TSR). The fraction of valid episodes that succeed:

$$\text{TSR} = \frac{|\mathcal{E}_s|}{|\mathcal{E}|}, \quad R_{\text{TSR}} = 100 \cdot \text{TSR}. \quad (5)$$

The good and bad thresholds are 1.0 and 0.0, respectively; a TSR of zero also caps dexterity at 35 so that a policy that never completes the task cannot receive a high score.

Task Completion Time. Mean wall-clock duration over successful episodes,

$$\bar{T} = \frac{1}{|\mathcal{E}_s|} \sum_{i \in \mathcal{E}_s} T_i, \quad (6)$$

normalized with $T_{\text{good}} = 0.35 T_{\max}$, an efficient completion well inside the timeout, and $T_{\text{bad}} = T_{\max}$. The metric is not applicable when $|\mathcal{E}_s| = 0$.

Comfort-Aware Path Efficiency ($\bar{\eta}_c$). This metric compares the path with a comfort-respecting reference path L_{out}^* computed outside bystander comfort zones, so that a socially necessary detour is not scored as inefficiency:

$$\bar{\eta}_c = \frac{1}{|\mathcal{E}_s|} \sum_{i \in \mathcal{E}_s} \text{clip} \left(\frac{L_{\text{out},i}^*}{\max(L_{\text{out},i}, \varepsilon)} \right). \quad (7)$$

The metric is computed on the approach prefix with the target excluded from bystander comfort tests; if less than 10% of the actual path lies outside comfort bands, the per-episode value is set to zero. Its thresholds are 1.0 (good) and 0.35 (bad).

Hesitation Rate (HR). Because unintended stops and reversals suggest indecision, HR counts stop-and-go epochs (speed below 0.05 m s^{-1} for at least 0.5 s, excluding social yielding) and displacement-gated reversals per unit episode time. The good and bad thresholds are 0 and 0.5 events/s, respectively, with lower values preferred.

Perceived safety. Perceived safety captures whether the robot respects personal space, with thresholds from Hall's proxemic zones [14]. Let $d_j(t) = \|\mathbf{p}_r(t) - \mathbf{p}_{h,j}(t)\|_2$ be the distance to the j -th person and $d(t) = \min_j d_j(t)$.

Minimum Human-Robot Distance ($d_{\min}$).

$$d_{\min} = \min_t d(t). \quad (8)$$

Good is the personal-space boundary $d_{\text{ps}} = 1.2$ m; bad is the hard floor $d_{\text{hard}} = 0.45$ m, below which contact is imminent.

Proxemic Intrusion Dose (D_{prox}). Discomfort grows with both depth and duration of intrusion, so the dose integrates penetration below the personal-space boundary over time, in m s:

$$D_{\text{prox}} = \int_0^T \max(0, d_{\text{ps}} - d(t)) dt. \quad (9)$$

The good threshold is 0, while the bad threshold $0.25 d_{\text{ps}} T$ corresponds to a quarter-episode at full-depth intrusion.

Speed Near Humans P95 (v_{95}). To capture fast close passes that an average can hide, we compute the 95th percentile of robot speed while within $d_{\text{near}} = 2.0$ m of any person. The good and bad thresholds are $\leq 0.25 \text{ m s}^{-1}$ and $\geq 1.0 \text{ m s}^{-1}$, respectively, and the feature is not applicable if the robot never enters the near-human zone.

Caps on this axis. A hard-distance violation caps safety at 25, while any collision, fall, or invalid action caps it at 20. A

high-risk close pass, meaning proxemic dose and near-human speed both in their bad ranges, caps safety at 35; when several caps fire, the lowest applies.

Perceived social awareness. This axis evaluates whether the robot interprets and responds to social cues. Gesture response, acknowledgment clarity, and human-aware approach are driven by structured *come-here* task events, each naming a target human, exposing the bearing error $\theta_e(t)$ between the robot's forward axis and the target direction, and defining an acknowledgment window $T_{\text{ack}} = 3\text{s}$; bystander acknowledgment is computed independently from passing events.

Gesture Response Success (GRS). This binary per-episode feature measures whether the robot responds to the correct person: $\text{GRS}_i = 1$ if the robot starts moving toward the designated target within T_{ack} without first approaching anyone else.

$$R_{\text{GRS}} = \frac{100}{|\mathcal{E}|} \sum_{i \in \mathcal{E}} \text{GRS}_i. \quad (10)$$

Acknowledgment Clarity (ACK). This feature records whether the robot faces the caller before approaching, indicating that it understood the request: $\text{ACK}_i = 1$ if the forward axis enters the cone $|\theta_e(t)| \leq 35^\circ$ within T_{ack} and before locomotion toward the target begins.

$$R_{\text{ACK}} = \frac{100}{|\mathcal{E}|} \sum_{i \in \mathcal{E}} \text{ACK}_i. \quad (11)$$

Acknowledgment clarity drives the largest social-awareness discrepancy discussed in Section VI.

Human-Aware Approach. The HA score combines target progress π_i , bystander clearance β_i , and a front-side approach angle α_i over the components exposed by the scenario:

$$\begin{aligned} C_i &= \{\pi_i, \beta_i, \alpha_i\}_{\text{app}}, \\ \text{HA}_i &= \text{mean}(C_i), \\ \pi_i &= \text{clip}\left(\frac{d_{0,i} - d_{f,i}}{d_{0,i}}, 0, 1\right), \end{aligned} \quad (12)$$

where $d_{0,i}$ and $d_{f,i}$ are initial and final robot-to-target distances, β_i is the fraction of episode time all bystanders stay outside personal space, and α_i scores approach bearing around the preferred $\pm 45^\circ$ front-side directions; the good and bad thresholds are ≥ 0.9 and ≤ 0.4 , respectively.

Bystander Acknowledgment (BA). This feature measures whether the robot maintains or increases clearance and slows down when passing a bystander. Per pass, a clearance-change term $\lambda = \text{clip}(0.5 + 0.5(d_{\text{pass}} - d_{\text{detect}})/d_{\text{ps}})$ and a slowdown term $\sigma = \text{clip}((v_{\text{before}} - v_{\text{inside}})/\max(v_{\text{before}}, \varepsilon))$ are blended:

$$\text{BA} = \overline{0.5\lambda + 0.5\sigma} \quad (\text{mean over passes}). \quad (13)$$

The good and bad thresholds are ≥ 0.5 and ≤ 0.05 , respectively, and the feature is not applicable if no bystander enters the near-human zone.

Impression. Impression covers motion quality and naturalness, independently of task outcome.

Motion Smoothness (SPARC). The SPARC index [43] integrates the arc length of the normalized speed spectrum $\hat{V}(\omega) = V(\omega)/V(0)$:

$$\text{SPARC} = - \int_0^{\omega_c} \sqrt{\omega_c^{-2} + \left(\frac{d\hat{V}}{d\omega}\right)^2} d\omega. \quad (14)$$

More negative values indicate jerkier motion, with good and bad thresholds of ≥ -2.0 and ≤ -6.0 , respectively.

Heading Jerk (HJ). This feature captures abrupt turns missed by translational SPARC by applying Eq. (14) to the yaw-rate signal $\dot{\psi}(t)$ with the same thresholds. In the present experiment it saturated at its bad threshold in every episode, as discussed in Section VII.

Stability (SC). This score combines instability events N_{inst} , unintended contacts N_{col} , and invalid actions N_{inv} :

$$\text{SC} = 1 - \text{clip}\left(\frac{N_{\text{inst}} + N_{\text{col}} + N_{\text{inv}}}{3}, 0, 1\right), \quad (15)$$

with $R_{\text{SC}} = 100 \cdot \text{SC}$; any such event caps safety at 20.

Legibility (L_{commit}). Committing early to a direction makes intent readable [44]. The commit time t_c is the first moment the robot stays within $\theta_c = 30^\circ$ of the target for at least 1 s; post-commit cone coverage f discounts trajectories that never settle:

$$L_{\text{commit}} = (1 - t_c/T) \cdot f. \quad (16)$$

The good and bad thresholds are ≥ 0.7 and ≤ 0.1 , respectively.

Naturalness (BN). This morphology-specific feature uses the normalized residual of a sinusoidal fit to a periodic joint signal for legged robots, and the ratio of lateral to forward RMS body-frame velocity for wheeled robots. The good threshold is 1.0 and the bad threshold depends on morphology; the feature requires a declared, supported morphology.

Feature-level results.

Table V gives the normalized feature scores behind the axis values of Table II, averaged over applicable runs and used in Section VI to interpret the results.

Three features explain most of the policy difference: task success rate separates the policies because Policy B times out in the moving-bystander scenario, and stability separates them because Policy B registers adverse events in twice as many runs. Acknowledgment clarity gives the largest difference, with Policy A satisfying it in every run and Policy B in one of six; its large social-awareness weight produces most of the axis gap that participants did not reproduce.

Two results reflect scenario coverage rather than policy differences: the proxemic features are applicable in only two runs per policy because only the moving-bystander scenario places people on the robot's path, and minimum clearance and human-aware approach are floored for both policies in those runs, while normalized heading-jerk scores are zero.

TABLE V
NORMALIZED FEATURE SCORES, AVERAGED OVER THE RUNS IN WHICH
EACH FEATURE APPLIED.

Feature	Policy A		Policy B	
	Score	<i>n</i>	Score	<i>n</i>
Task success rate	100.0	6	66.7	6
Task completion time	100.0	6	100.0	4
Comfort-aware path eff.	91.8	6	82.8	4
Hesitation rate	100.0	6	100.0	6
Minimum human-robot dist.	0.0	2	0.0	2
Proxemic intrusion dose	44.9	2	12.1	2
Speed near humans (P95)	0.0	2	67.5	2
Acknowledgment clarity	100.0	6	16.7	6
Human-aware approach	0.0	2	0.0	2
Motion smoothness (SPARC)	100.0	6	92.2	6
Heading jerk	0.0	6	0.0	6
Stability	66.7	6	33.3	6
Legibility	100.0	6	100.0	6

n is the number of applicable runs out of six per policy. Gesture response success, bystander acknowledgment, and behavioral naturalness were not applicable in any run of this scenario set and are omitted. Speed near humans favors Policy B, most likely because Policy B had stopped after timing out during part of the interval in which it was near a person.

APPENDIX B HUMAN-STUDY DETAILS

Camera viewpoint. The two views in Fig. 5 do not shift the overall rating level: every bootstrap confidence interval for the mean first-versus-third-person difference crosses zero. The third-person view widens the Policy A minus Policy B gap on all four axes, most clearly on safety (+0.614 Likert points, $p = 0.021$) and social awareness (+0.652, $p = 0.014$). The gaps (first/third person) are 0.886/1.243 for dexterity, $-0.243/0.371$ for safety, $-0.536/0.116$ for social awareness, and $0.414/0.514$ for impression. This may be a visibility effect: clearance and passing speed are easier to judge from a bystander’s position than from the position of the person being approached. The four tests are exploratory and uncorrected for multiplicity. Future protocols should therefore control viewpoint, although this study does not establish a general effect size.

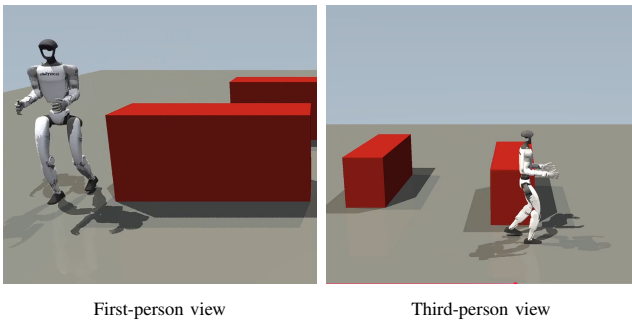


Fig. 5. First- and third-person views used in the human study.

Prior robot experience. Because this analysis compares rater groups, it uses the full cohort after excluding two responses with missing familiarity labels, leaving $n = 23$ participants with daily robot contact and $n = 95$ without.

Fig. 6 shows participant-level means: after averaging each participant’s scores across policies and axes, daily-contact participants rate 9.0 points lower on the $[0, 100]$ scale, with a bootstrap 95% interval of $[-16.5, -1.3]$. The gap is largest on social awareness (-10.0), then impression (-9.5), safety (-8.8), and dexterity (-7.8). Policy-specific shifts follow the same direction: -10.3 points for Policy A (95% interval $[-18.7, -1.6]$) and -7.7 for Policy B ($[-17.3, 2.3]$). This cross-sectional comparison shows that the daily-contact group gave lower ratings; it does not show that experience caused the difference. The affine alignment of Eq. (3) may not transfer across rater groups.

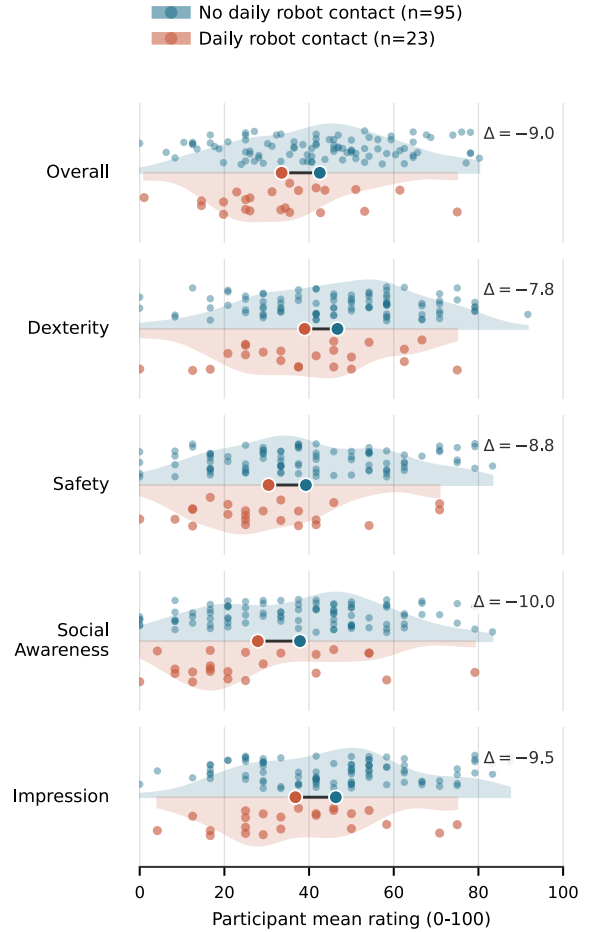


Fig. 6. Ratings by daily robot contact, averaged over policies and axes. Dots denote participants; black markers show group means and bootstrap 95% confidence intervals.

Study instrument.

This section documents the wording and response coding of the five-minute online survey. Recruitment was conducted mainly through Politecnico di Milano and the AEA student association, and English and Italian were shown together.

Protocol. The questionnaire had three parts: participants read the introduction and gave consent, answered one background item about robotics familiarity, and then rated four short

videos using the same four items. The four videos covered both policies and both camera viewpoints, in a fixed order, and were presented without any label identifying the policy: participants were never told that two different controllers were being compared, or which video showed which. No other data, including personal data beyond the familiarity item, were collected.

Survey introduction

Human-Robot Interaction Evaluation [~ 5 minutes]

This survey is part of the research paper “AsimovBM: Toward Automatic Evaluation of Human-Perceived Traits in Physical AI Robots.” You will watch 4 short robot interaction videos. After each video, please answer a few questions about your perception of the robot’s behavior.

Valutazione Human-Robot Interaction [~ 5 minuti].

Questo sondaggio fa parte della ricerca per il paper “AsimovBM: Toward Automatic Evaluation of Human-Perceived Traits in Physical AI Robots.” Vedrai 4 brevi video di interazione robotica. Dopo ogni video, rispondi ad alcune domande sulla tua percezione del comportamento del robot.

Background item. The background item had two response options defining the groups analyzed in Appendix B.

Background / Familiarity with Robotics (*Esperienza/Familiarità con la robotica*)

- ☐ No daily contact with robots
Nessun contatto quotidiano con robot
- ☐ Daily interaction with robots (e.g., robotics student or professional worker)
Interazione quotidiana con robot, ad esempio come studente di robotica o professionista del settore

Rating items. The same four items were repeated after each video, numbered Q1.1–Q1.4 through Q4.1–Q4.4, with each item mapped to one axis of Section III. The arrows below are paper annotations; participants did not see the axis names or the metric framework.

Rating items, repeated for each of the four videos

- Qn.1** The robot seemed competent and in control. $\rightarrow \mathcal{D}$
Il robot sembrava competente e sotto controllo.
- Qn.2** The robot’s behavior felt safe around people or obstacles. $\rightarrow \mathcal{S}$
Il comportamento del robot sembrava sicuro in presenza di persone o ostacoli.
- Qn.3** The robot seemed aware of its surroundings, including people or obstacles. $\rightarrow \mathcal{A}$
Il robot sembrava consapevole dell’ambiente circostante, incluse persone e ostacoli.
- Qn.4** Overall, I had a positive impression of the robot. $\rightarrow \mathcal{I}$
Nel complesso, ho avuto un’impressione positiva del robot.

Response scale and implementation. Every item used the same fully labeled, symmetric 7-point agreement scale.

Response scale (identical for all items)

- −3 Strongly disagree *Fortemente in disaccordo*
- −2 Disagree *In disaccordo*
- −1 Slightly disagree *Leggermente in disaccordo*
- 0 Neutral *Neutrale*
- +1 Slightly agree *Leggermente d’accordo*
- +2 Agree *D’accordo*
- +3 Strongly agree *Fortemente d’accordo*

A raw response $v \in \{-3, \dots, +3\}$ is mapped to the $[0, 100]$ scale used throughout the paper by

$$h = 100 \cdot \frac{v + 3}{6}, \quad (17)$$

so that “strongly disagree” becomes 0, “neutral” becomes 50, and “strongly agree” becomes 100. Because one Likert step equals 16.7 points, the viewpoint effects near 0.6 Likert points in Appendix B equal about 10 points after rescaling.

For each participant, the score on axis k for a policy is the mean of Eq. (17) over the two videos showing that policy, and policy comparisons are paired because every participant rated both. Participation was voluntary, anonymous, and uncompensated, with consent obtained before the first video.